NIRANJAN P

niranjanprabakar325@gmail.com
linkedin.com/in/niranjan-p-497450278 | github.com/NiranjanPrabakar | Google Scholar | niranjanprabakar.github.io

RESEARCH SUMMARY

Computer Science graduate specializing in Artificial Intelligence and Robotics, with research experience in dual arm manipulation, perception guided planning, and autonomous navigation. My work centers on generalizable robot manipulation, focusing on Vision Language Action architectures that remain robust under unseen workspace conditions, combining deep learning, computer vision, SLAM, and classical robotics methods within ROS2 based systems. This experience spans designing, developing, and deploying end to end intelligent systems whose competence extends reliably beyond controlled training conditions.

EDUCATION

Vellore Institute of Technology, Chennai, India 2022 to 2026
Bachelor of Technology in Computer Science (Specialization: AI & Robotics) *CGPA: 9.15/10.0*

Chettinad Vidyashram, Chennai, India 2021 to 2022
Higher Secondary Education (Grade 12) *Score: 91%*

Chettinad Vidyashram, Chennai, India 2019 to 2020
Secondary Education (Grade 10) *Score: 95.6%*

RESEARCH EXPERIENCE

Research Officer (Statistical) Jul 2026 to Present
*A*STAR, Institute of Advanced Intelligence and Computing (IAIC)* *Singapore*

- Curated a four camera RGB and depth bimanual manipulation dataset with synchronized proprioception for pick, place, and label reading tasks
- Fine tuned and validated VLA policies on collected demonstrations, validating corrected models on physical hardware
- Developed a cross arm data augmentation technique through joint and video mirroring, extending training coverage
- Research in progress: designing a dual arm VLA architecture with depth grounded relative pose conditioning, SAM based visual gating, and confidence triggered correction for generalization to unseen workspaces

Visiting Scholar Dec 2025 to Jun 2026
SUTD, Robotics and Automation Research (ROAR) Laboratory *Singapore*
Towards Autonomous Bed Making: Perception Guided VLM Planning for Mobile Manipulation
1st Author: Manuscript in preparation

- Designed a VLM planning system guided by perception for autonomous bed making on a mobile manipulator
- Built PGVP, cutting VLM spatial hallucinations by 92.8% and improving object recall by +34.9 pp
- Built a causal step learning engine reaching 97.5% step accuracy across 81 demonstration episodes with zero hand written definitions
- Trained an 11 class YOLOv8 and SAM segmentation model on 5,562 annotated images (mAP50 0.855) and a Depth Fusion module (92.5% accuracy)
- Validated on a Kinova Gen3 across 4 obstacle scenes with 100% task completion, zero intervention

Visiting Scholar May 2025 to Jul 2025
SUTD, Robotics and Automation Research (ROAR) Laboratory *Singapore*

- Built a FastAPI backend and ROS2 topic/service nodes for actuator control, syncing live state via Redis with the PLC's Modbus I/O process
- Built a responsive frontend and backend interface for remote operation with multiple view RealSense D457 streaming
- Delivered a scalable teleoperation platform enabling remote cleaning, spraying, washing, and scrubbing control

Visiting Scholar Jun 2024 to Jul 2024
SUTD, Robotics and Automation Research (ROAR) Laboratory *Singapore*
Online Coverage Path Planning in LIDAR Perception Constrained Environments [Link]
2nd Author: Published, IEEE TENCON 2024 Singapore

- Labeled, annotated, and curated a 4,303 image dataset targeting obstacle classes (glass, wires, sockets, carpet) undetectable by LiDAR due to range and reflectivity limitations
- Trained and benchmarked YOLOv8, RetinaNet, and Faster R-CNN, with the top performing model (84% precision, 78.78% mAP@0.5) selected for deployment

RESEARCH PROJECTS

Real Time Hand Gesture Recognition for Quadruped Robot Control: Stream Level Evaluation of Temporal Modeling Architectures and Command Safety Jul 2025 to Present [Link]
1st Author: Presented; under review, Journal of Real-Time Image Processing (Springer)

- Built an 18 clip, 6,975 frame video corpus with a public evaluation protocol
- Compared 9 architectures behind MediaPipe detection; static perfect models retained only 58 to 90% accuracy on real streams
- Identified a pass through failure across all 9 architectures, matching CNN and CNN LSTM performance (p>0.05)
- Designed a cross validated 667 ms dwell gate cutting wrong commands to 1 residual failure; a state space model matched its latency at 75% fewer parameters

Autonomous Search and Rescue Robot with Vision Based Object Detection and SLAM-Enabled Navigation in Hazardous Environments Jul 2025 to Nov 2025 [Link]
2nd Author: Published, EPJ Web of Conferences, vol. 367, art. 01011, RIACT 2026

- Trained YOLOv8n for fire and smoke detection (mAP50 0.84, precision 0.87, recall 0.83) on a 3,000 image dataset
- Developed HSV victim detection with a coordinate transformation pipeline mapping detections to the global map frame
- Integrated SLAM Toolbox localization (4.2 cm mean pose error) on a TurtleBot3 Waffle platform
- Designed a deterministic frontier exploration algorithm achieving 97.3% map coverage in 8.4 minutes and 91.3% Nav2 navigation success, outperforming baselines (Gazebo simulation)

Enhanced Composite Deterministic Finite Automaton with Deep Neural Segmentation and Vision-Language Synthesis for Brain Tumor Progression Analysis and Risk Stratification Jul 2025 to Nov 2025 [Link]
1st Author: Presented at IEEE IECS 2026, VIT Bhopal, India; manuscript submitted, publication pending

- Trained a YOLOv12 segmentation model (93.81% mAP@50, 90.76% precision, 88.0% recall)
- Designed a 19 state DFA over 6 morphological features for tumor progression modeling
- Layered Bayesian risk stratification, reaching 99.3% confidence on a normal brain control case
- Built a treatment response and surgical candidacy module, synthesized by Gemini 2.5 Flash into structured 7 section clinical reports

Enhanced Deep Ensemble Framework for High-Resolution Rainfall Forecasting: Multi-Scale Feature Integration and Uncertainty Quantification Dec 2024 to Apr 2025 [Link]
1st Author: Published, IEEE ICSCSS 2025, Coimbatore, India

- Designed an attention enhanced LSTM with adaptive gradient boosting fusion, dynamically weighting ensemble members by prediction confidence
- Developed uncertainty quantification using Monte Carlo dropout and ensemble variance
- Engineered multiple scale temporal features from 115 years of IMD climate data
- Achieved 23.4% RMSE reduction, 18.7% MAE improvement, and 87.3% reliability score

RGB, Depth, and Thermal Fusion for Autonomous Fruit Ripeness Detection and Selective Harvesting Nov 2025 to Present
Manuscript in preparation

- Built a dual branch RGB and depth YOLOv8n detector with a downstream thermal classifier for ripeness and defect grading across apple, banana, and mango
- Achieved 95.8% mAP@0.5, +6.8 pp over RGB only and +3.2 pp over RGB and depth baselines
- Validated the thermal ripeness signal via a 5 stage study showing a climacteric temperature peak
- Integrated with a 6 DOF manipulator (ROS2/MoveIt2): 92% success, 100% selectivity in Gazebo

ESP32 Based WiFi Controlled Web Interface for a 6 DOF Robotic Arm Jul 2024 to Nov 2024 [Link]

- Built real time bidirectional control over the arm via AsyncWebSocket
- Implemented record and playback automation with timestamp based motion sequencing
- Built PWM servo control architecture across a 0 to 180° range

TECHNICAL SKILLS

Programming Languages:	Python, JavaScript, MATLAB, PL/SQL
Robotics & AI:	ROS2, Gazebo, Nav2, SLAM, Computer Vision, OpenCV, MediaPipe
Machine Learning:	LSTM, Gradient Boosting, Random Forest, Bayesian Networks, Vision Language Models
Embedded Systems:	Arduino, ESP32, IoT, Servo Control, Sensor Integration
Tools & Frameworks:	React.js, HTML/CSS, AWS